

ONKAR BANSODE

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Technical Skills

Programming Languages: Python, C++, C, SQL
AI / Machine Learning: Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, XGBoost
Deep Learning: PyTorch, TensorFlow, Keras, CNN (ResNet50), ANN
Modern AI: Generative AI, LLMs, Prompt Engineering, RAG, Agentic AI
Backend Development: Flask, FastAPI, REST APIs
Databases: MySQL, MongoDB
Software Engineering: DSA, OOP, SDLC, Agile Methodologies, Authentication
Developer Tools: Git, GitHub, Postman, Jupyter Notebook

Education

Government College of Engineering & Research, Avasari Kh., Pune	2023–2026
Bachelor of Engineering (Computer Engineering)	CGPA: 9.49/10.00
Government Polytechnic, Solapur	2020–2023
Diploma in Computer Science & Engineering	90.46%

Experience

1) Project Intern NCRA–TIFR (GMRT)	Jan 2026 – Apr 2026
<i>Python, Flask, REST APIs, TCP/IP</i>	
- Developed a Python-based RF control system enabling remote configuration of 60 RF channels. - Built modular REST APIs with hardware abstraction and automated device discovery. - Contributed to RFI Detection and Switchable Notch Filter Bank systems.	
2) Full Stack Developer Intern Maveric InfoTech	Jul 2025 – Aug 2025
- Developed backend modules and worked on API integration and database systems. - Gained experience in debugging, SDLC workflows and team-based development.	

Projects

1) KrushiMitra – AI-Powered Agricultural Companion	GitHub
<i>TypeScript, React, Python, FastAPI, PyTorch, ResNet50, Tailwind CSS</i>	
- Engineered a production-ready AI platform delivering crop disease detection, weather insights, and personalized farming recommendations, improving decision-making for end users. - Deployed a fine-tuned ResNet50 transfer learning model using PyTorch for real-time plant disease classification from image inputs with high inference efficiency. - Architected an asynchronous FastAPI backend with secure authentication (bcrypt) and optimized API response handling for scalable model serving. - Designed and developed a responsive SPA using React, TypeScript, Vite, and Tailwind CSS, ensuring accessibility for non-technical users. - Implemented intelligent AI agents to process contextual agricultural data and generate real-time, actionable advisory outputs.	
2) Criticality Prediction for Organ Transplantation & Disease Diagnosis (Published Research)	GitHub
<i>Python, TensorFlow, Scikit-learn, Flask, Pandas, NumPy</i>	
- Developed a multi-organ AI diagnostic system to classify disease severity (mild, moderate, severe) across Heart, Lungs, Kidney, and Liver for clinical decision support. - Trained and benchmarked ML/DL models including CNNs, Random Forest, and XGBoost on medical datasets (UCI, PhysioNet), optimizing performance using precision, recall, and ROC-AUC. - Engineered robust data preprocessing pipelines handling missing clinical attributes, normalization, and feature engineering for improved model accuracy. - Built and deployed a Flask-based web application enabling real-time patient risk prediction through REST APIs and interactive frontend. - Structured modular architecture for organ-specific models, enabling scalability and future integration of additional diagnostic modules.	

Leadership & Achievements

- Selected for **NCRA–TIFR Students Training Programme (STP-2025)**.
- 1st Prize**, National Slogan Writing Competition – Government of India.
- Vice Chairperson & Technical Head**, CSAC (2023–2026).
- Core Member, COMPSA (2024–2026).